AP CALCULUS AB – UNIT 4

4.4 Related Rates

Name: ____________ **Class:** ____________ **Date:** ____________

Lesson Overview and Learning Objectives

- Identify changing quantities linked by a geometric or physical constraint.
- Differentiate a constraint with respect to time using implicit differentiation.
- Substitute information from a specified instant and solve for an unknown rate.
- Interpret the sign and units of the resulting rate.

Keys: Essential Knowledge and Vocabulary

Related rates	Rates of two or more quantities connected by an equation and changing with a common variable, usually time.
Constraint	An equation relating the changing quantities, such as $x^2 + y^2 = z^2$ or $V = \frac{1}{3}\pi r^2 h$.
Differentiate first	Keep variables symbolic while differentiating; substitute the instant afterward.
Chain Rule	$\frac{d}{dt}(x^n) = nx^{n-1}x'$ because $x = x(t)$.
Sign convention	Choose positive directions first; a negative answer means change opposite the chosen positive direction.
Similarity	In cone and shadow problems, use similar triangles to eliminate an extra variable before differentiating.

Explanation, Strategy, and Common Errors

- Draw and label a diagram; identify constants, variables, known rates, and the requested rate.
- Write one constraint containing the target variable and the variables whose rates are known.
- Differentiate every changing quantity with respect to the same independent variable.
- Only after differentiation, substitute the values at the specified instant and solve.
- State the final rate with sign, units, and a complete contextual interpretation.

Solved Example: Expanding circle

A circle's radius increases at 3 cm/s. How fast is its area increasing when $r = 5$ cm?

Answer and Reasoning

$A = \pi r^2$, so $A' = 2\pi r r' = 2\pi(5)(3) = 30\pi$ cm^2/s.

Solved Example: Ladder

A 10-m ladder rests against a wall. The base moves away at 0.6 m/s. Find the top's rate when the base is 6 m from the wall.

Answer and Reasoning

$x^2 + y^2 = 100$. At $x = 6$, $y = 8$. Then $2xx' + 2yy' = 0$, so $y' = -\frac{6(0.6)}{8} = -0.45$ m/s.

Solved Example: Conical tank

Water in a cone satisfies $r/h = 1/3$. If $h' = 0.2$ m/min, find V' when $h = 6$ m.

Answer and Reasoning

$r = h/3$, so $V = \frac{1}{3}\pi(h^2/9)h = \frac{\pi}{27}h^3$. Thus $V' = \frac{\pi}{9}h^2h' = \frac{\pi}{9}(36)(0.2) = 0.8\pi$ m^3/min.

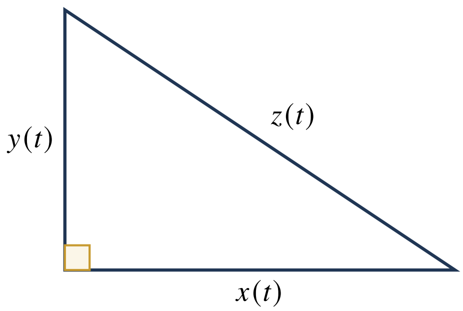
Solved Example: Approaching vehicles

One vehicle is 3 km east of an intersection moving east at 40 km/h; another is 4 km north moving north at 30 km/h. Find the rate of change of their separation.

Answer and Reasoning

$z^2 = x^2 + y^2$. $z = 5$, so $z' = (3 \cdot 40 + 4 \cdot 30)/5 = 48$ km/h; separation is increasing.

Shared Representation / Data



$x^2 + y^2 = z^2$
Differentiate with respect to t before substituting an instant.

AP Exam Language

For every applied derivative, include the sign, units, input value, and a complete interpretation. In related-rates work, differentiate before substituting. For LĤopital problems, verify an allowed indeterminate form first.

Leveled Practice

Easy Foundations

Build fluency with the definition, notation, and one-step applications.

1. Differentiate $x^2 + y^2 = 25$ with respect to t.

2. Differentiate $V = \frac{4}{3}\pi r^3$ with respect to time.

3. A square side grows at 2 cm/s. Find A' when side is 7 cm.

4. A sphere radius decreases at 0.4 cm/min. State the sign of V'.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. A circle area increases at 12π cm^2/s. Find r' when $r = 3$ cm.

6. A cube volume increases at 54 cm^3/s. Find the side rate when $s = 3$ cm.

7. A 13-ft ladder has base 5 ft from a wall and moving away at 2 ft/s. Find y'.

8. For $xy = 60$, find y' when $x = 5$, $y = 12$, and $x' = 3$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Air enters a sphere at 100 cm^3/s. Find r' when $r = 4$ cm.

10. A person 1.8 m tall walks away from a 6-m lamp at 1.5 m/s. Find the shadow-tip speed.

11. Water fills a cylindrical tank of radius 2 m at 3 m^3/min. Find h'.

12. Two sides of a rectangle satisfy $x' = 2$, $y' = -1$. When $x = 8$, $y = 5$, find perimeter and area rates.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. A cone has fixed height 12 cm and changing radius. If $V' = 20\pi$ cm^3/s when $r = 5$, find r'.

14. A point moves on $xy = 24$. At $(6, 4)$, $x' = 3$. Find y'' if $x'' = 0$.

15. A camera 20 m from a road tracks a car moving at 15 m/s. If θ is the viewing angle and the car is 20 m along the road, find θ'.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. A right circular cone preserves volume while its height increases. Derive the relation between r'/r and h'/h.

17. A particle on $x^2 + 4y^2 = 100$ has $x' = 5$ at $(6, 4)$. Find y' and interpret its sign.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. In a related-rates problem, numerical values should usually be substituted

(A) before writing a constraint
(B) before differentiating
(C) after differentiating
(D) only after solving the entire problem

2. If $A = \pi r^2$, then

(A) $A' = 2\pi r$
(B) $A' = 2\pi r r'$
(C) $A' = \pi (r')^2$
(D) $A' = \pi r^2 r'$

3. For $x^2 + y^2 = 100$, if $x > 0, y > 0, x' > 0$, then y' is

(A) positive
(B) negative
(C) zero
(D) not determined

4. A negative answer for an unknown rate means

(A) the calculation is wrong
(B) the quantity is negative
(C) the quantity changes opposite the chosen positive direction
(D) time is decreasing

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. A sphere's volume increases at 75 cm^3/s. At radius 3.2 cm, r' is closest to

(A) 0.29
(B) 0.58
(C) 1.83
(D) 7.32

6. A 15-m ladder base is 9 m from a wall and moves away at 1.2 m/s. The top rate is

(A) −0.90
(B) −0.72
(C) 0.72
(D) 0.90

AP-Style Free-Response Practice

No Calculator

Water is poured into an inverted cone of height 12 m and top radius 4 m. The water depth is h and surface radius is r.

(a) Use similarity to express r in terms of h and write volume as a function of h. [2 points]

(b) If water enters at 2 m^3/min, find h' when $h = 6$. [3 points]

(c) At that instant find r'. [2 points]

Calculator Permitted

Two drones fly at the same altitude from a common point. Drone A travels east at 24 m/s and Drone B travels north at 18 m/s.

(a) Write a relationship for their separation z after t seconds. [2 points]

(b) Find z' when $t = 10$. [3 points]

(c) Explain why the answer is constant for all $t > 0$. [2 points]